

Vibe Index: Roaring Bitmap Positional Phrase Matching for LLM Context Retrieval

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Abstract

Large Language Model applications using Retrieval-Augmented Generation (RAG) pipelines waste significant computational resources by injecting entire retrieved document chunks into context windows, despite relevant content occupying only a small fraction of each chunk. I present Vibe Index, a sub-microsecond exact phrase matching engine using Roaring Bitmaps and anchor-and-offset positional scanning. My approach achieves 112 nanosecond latency for exact phrase matching on single-match queries and 1.83 milliseconds for indexing 50K tokens — up to four orders of magnitude faster than embedding-based semantic search. The system provides exact token positions (not document-level matches) using approximately 0.5 MB of memory for 50K tokens, enabling injection of only ± 50 tokens around matches rather than 1K–4K token chunks, reducing token consumption by 93% and saving approximately 5.65 MB of KV cache VRAM per query (cumulative savings of ~5.5 GB across 1000 queries) for 7B parameter models. I evaluate the system on real Rust source code and demonstrate the algorithmic foundation for hybrid retrieval combining BM25 document-level candidate selection with exact positional validation.

1. Introduction

Modern LLM applications increasingly rely on Retrieval-Augmented Generation (RAG) pipelines to ground model outputs in external knowledge. The standard RAG retrieval paradigm operates as follows: (1) a retriever — typically embedding-based semantic search or BM25 keyword matching — identifies relevant document chunks from a corpus; (2) the entire retrieved chunk (typically 1,000–4,000 tokens) is injected into the LLM's context window; (3) the model generates a response conditioned on this context.

This paradigm contains a fundamental inefficiency. Retrievers identify which chunks are relevant but cannot determine where within those chunks the relevant content resides. The actual answer or code pattern may occupy only 2–3 lines (approximately 50–100 tokens), yet the full chunk is injected. This wastes:

- * Token budget: Each unnecessary token costs money and consumes context window space
- * KV cache: A 7B parameter model processes approximately 1.5 KB per token in KV cache. Injecting 4,096 tokens instead of 300 relevant tokens wastes approximately 5.65 MB of VRAM per query
- * Inference latency: Each additional token adds approximately 30+ seconds of generation time for typical decoder-only models

* Output quality: Irrelevant context increases the probability of model hallucination or distraction

Existing solutions address document-level retrieval but not positional precision within documents. Embedding search provides semantic relevance at millisecond latency. BM25 provides keyword matching at microsecond latency but returns document-level matches without positional information. Vector databases (FAISS, Chroma, Weaviate) provide semantic search but operate at 5–20 ms latency with 20+ MB memory footprint for 50K tokens.

I propose a complementary approach: Vibe Index, a positional phrase matching engine that finds exact phrases at exact token positions in sub-microsecond time. Rather than replacing semantic retrieval, Vibe Index operates as a second-stage refinement: embeddings or BM25 identify candidate chunks, and Vibe Index pinpoints the exact lines within those chunks where relevant content resides.

Contributions

- I present an anchor-and-offset phrase matching algorithm using Roaring Bitmaps that achieves 112 ns latency for single-match queries

- I describe a bigram prefiltering strategy for fuzzy search that reduces Levenshtein distance computations by 97%

- I demonstrate file-aware indexing with line-level granularity, FNV-1a content hashing for incremental updates, and hot/cold storage for scalable indexing

- I provide empirical evaluation on 50K synthetic tokens and real Rust source code, comparing against BM25 and embedding-based baselines

- I demonstrate practical integration with LLM backends (llama.cpp, vLLM) showing 93% token reduction and estimated 5.65 MB KV cache VRAM savings per query

2. Related Work

2.1 Dense Retrieval

Dense retrieval using neural embeddings (DPR [1], ANCE [2], E5 [3]) has become the dominant approach for document retrieval in RAG pipelines. Embedding-based methods achieve semantic matching capability — "login" matches "authentication" — but operate at 5–20 ms latency and provide only document-level relevance scores without positional information. The FAISS library [4] provides efficient approximate nearest neighbor search but requires approximately 20 MB of memory for 50K tokens and cannot identify exact token positions within documents.

2.2 Sparse Retrieval

BM25 [5] remains the standard sparse retrieval algorithm, used in Elasticsearch, Lucene, and Tantivy. BM25 provides document-level scoring based on term frequency-inverse document frequency, operating at 50–500 μ s latency. While BM25 can return term positions within documents, it does not support efficient multi-term phrase matching at sub-microsecond latency. My hybrid search module

(Section 5.4) uses BM25 for candidate document selection and Vibe Index for exact position validation within candidates.

2.3 Exact Phrase Matching

Traditional exact phrase matching operates via linear scan ($O(N)$) or inverted index with position lists. Modern search engines (Lucene, Tantivy) use position lists encoded as variable-length integers, requiring intersection of position lists for multi-term phrases. Roaring Bitmaps [6] provide compact, efficiently-operations position encoding. My contribution is applying Roaring Bitmaps specifically to the LLM context retrieval problem with anchor-and-offset optimization targeting sub-microsecond latency.

2.4 Fuzzy Matching

Fuzzy string matching using Levenshtein distance [7] is computationally expensive: $O(m \times n)$ for strings of length m and n . Bigram-based prefiltering [8] reduces candidate set size by looking up substrings, dramatically reducing the number of full Levenshtein computations. I adapt and optimize this technique for token-level fuzzy matching with configurable edit distance.

3. System Architecture

3.1 Overview

Vibe Index consists of the following components:

- * TokenLexicon: Bidirectional mapping between token strings and u32 identifiers, with bigram index for fuzzy prefiltering
- * Roaring Bitmap Index: One bitmap per unique token, storing positions where the token appears
- * FileIndex: File-aware tracking with path, content, line offsets, token-to-line mapping, and FNV-1a content hashing
- * QueryParser: Natural language to search phrase conversion, handling code syntax (camelCase, snake_case, `` paths, generics)
- * HotColdStorage: In-memory hot layer with disk-backed cold segments for scalable indexing
- * PromptInjector: Context window builder that extracts relevant windows around matches

The core data flow is:

```
Token stream → TokenLexicon (u32 IDs) → Roaring Bitmap per token →  
Anchor-and-offset phrase scan → MatchResult[] with file/line metadata
```

3.2 Token Lexicon

The TokenLexicon maintains a bidirectional mapping between token strings and compact u32 identifiers. This serves two purposes: (1) memory efficiency — u32 values are 4 bytes each versus

variable-length strings, and (2) enabling bitmap-based operations on integer positions.

For each unique token, a u32 ID is assigned and stored in a hash map. The inverse mapping (u32 ID → string) is maintained for result presentation. The lexicon also maintains a bigram index: for each token, all overlapping character bigrams are extracted and mapped to the token's u32 ID. This bigram index enables efficient fuzzy candidate generation (Section 4.2).

3.3 File-Aware Indexing

Unlike traditional search indexes that treat documents as flat token streams, Vibe Index maintains file-level metadata:

- * FileSegment: Each indexed file is represented as a `FileSegment` containing the file path, content, token range (start/end position in the global position stream), line offset map (token position → line number), and FNV-1a content hash
- * Line tracking: For each token added via `add\_file()`, the line number is determined by scanning content for newline characters and maintaining a running offset map
- * Content hashing: FNV-1a hash enables change detection for incremental updates without re-parsing unchanged files

This design enables results that include `file_path`, `line_number`, and `line_content` — essential for code search where developers need to know exactly which line contains a pattern.

4. Core Algorithms

4.1 Anchor-and-Offset Phrase Matching

Given a phrase query with tokens $[t_1, t_2, \dots, t_k]$, the goal is to find all positions P where all k tokens appear consecutively: $\text{position}(P) = P$, $\text{position}(P+1) = P+1$, ..., $\text{position}(P+k-1) = P+k-1$.

A naive approach would intersect k position lists, which is $O(k \times N)$ in the worst case. My anchor-and-offset algorithm reduces this to $O(\text{min_cardinality} \times k)$:

```
Algorithm AnchorAndOffset(phrase_tokens):
  For each token t_i in phrase:
    bitmap_i = lexicon.bitmaps[token_id(t_i)]

  anchor_bitmap = bitmap with minimum cardinality
  anchor_index = index of anchor token in phrase

  results = empty set

  For each position P in anchor_bitmap:
    match = true
    For offset j from 0 to phrase_length - 1:
      if j == anchor_index: continue
      expected_position = P + (j - anchor_index)
      if expected_position not in bitmap[phrase[j]]:
        match = false
        break
```

```

        if match:
            results.add(P)

    return results

```

Key insight: By selecting the token with the fewest occurrences as the anchor, I minimize the number of positions to check. For each anchor position, I verify that sibling tokens exist at the expected offset positions. This is particularly efficient for code search where common tokens like `fn`, `let`, `if` may appear frequently, but specific function names like `authenticate` or `process_request` appear rarely.

Complexity: $O(\min_cardinality \times \text{phrase_length})$ vs. $O(N \times \text{phrase_length})$ for linear scan.

Example: For query `["fn", "authenticate", "("]`:

- * ``fn`` appears 847 times → bitmap cardinality = 847
- * ``authenticate`` appears 3 times → bitmap cardinality = 3
- * ``(`` appears 1,204 times → bitmap cardinality = 1,204

Anchor = `authenticate` (cardinality 3). I check only 3 positions, verifying `fn` at P-1 and `(` at P+1. Total checks: $3 \times 2 = 6$ bitmap lookups.

4.2 Fuzzy Search with Bigram Prefiltering

Fuzzy search finds tokens within a specified Levenshtein distance of the query. A naive approach computes Levenshtein distance against all unique tokens, which is expensive for large vocabularies (672+ unique tokens in my test corpus).

Bigram prefiltering reduces the candidate set dramatically:

```

Algorithm FuzzySearch(query, max_distance):
    bigrams = extract_all_bigrams(query)
    candidates = empty set

    For each bigram in bigrams:
        candidate_ids = bigram_index[bigram]
        candidates = candidates ∪ candidate_ids

    // Length filter: skip tokens where |len(query) - len(token)| > max_distance
    filtered = empty set
    For each token_id in candidates:
        token = lexicon[token_id]
        if |len(query) - len(token)| <= max_distance:
            filtered.add(token_id)

    // Full Levenshtein only on filtered candidates
    results = empty set
    For each token_id in filtered:
        token = lexicon[token_id]
        if levenshtein(query, token) <= max_distance:
            results.add(token_id)

    return results

```

Example: Query `"proces"` (missing `'s'`):

- * Bigrams: `["pr", "ro", "oc", "ce", "es"]`

- * Candidates from bigram lookup: ~20 tokens (3% of 672 unique tokens)
- * After Levenshtein: ["process"] matches with distance 1
- * Without prefiltering: 672 Levenshtein computations
- * With prefiltering: ~20 Levenshtein computations (97% reduction)

Latency: 4.64 μ s for 1-character typo, 67 μ s for 2-character typo.

4.3 Unified Natural Language Search

The `search()` method bridges natural language queries and exact phrase matching:

Query parsing: `query_parser` splits natural language into search phrases, handling code syntax (camelCase splitting: "processRequest" $\rightarrow$ ["process", "Request"]), stop word removal ("where is the" $\rightarrow$ []), and Rust path syntax ("std::io::Read" $\rightarrow$ ["std", "io", "Read"])

Phrase search: Each parsed phrase is searched using anchor-and-offset

Fuzzy search: Each parsed phrase is also searched with fuzzy matching (`max_distance=1`)

Merge and rank: Results are merged, deduplicated by (file_path, line_number), and ranked by confidence score:

```
confidence = base_confidence * log(1 + file_size_tokens)

base_confidence:
- Exact phrase match: 0.95
- Fuzzy match: 0.50

file_size_weighting:
- Larger files get logarithmic boost (diminishing returns)
- Prevents small files from dominating results
```

5. Implementation Details

5.1 Roaring Bitmaps

I use the `roaring crate` (v0.10) for bitmap operations. Roaring Bitmaps provide:

- * Compression: Run-length encoding for sequences of consecutive integers (common in token position streams)
- * Fast intersection: SIMD-optimized bitmap AND operations
- * Memory efficiency: ~0.5 MB for 50K token positions across 100 unique tokens

For phrase search, I use `select_min` to find the bitmap with minimum cardinality (the anchor), then iterate over its positions and check offset positions in other bitmaps using `contains()`.

5.2 Hot/Cold Storage

For scalable indexing, Vibe Index implements a two-tier storage strategy:

- * Hot layer: In-memory buffer of recent tokens (configurable `max\_hot\_tokens`). All additions and searches operate here first
- * Cold layer: Disk-backed segments created when hot layer exceeds capacity. Each segment stores gzip-compressed token sequences (bincode) and base64-encoded Roaring Bitmaps

Cross-layer search (phrases spanning hot and cold boundaries) is supported by combining results from both layers. This design enables indexing of large codebases (100K+ tokens) while maintaining sub-microsecond query latency for the hot portion.

5.3 Persistence

Persistent storage uses a versioned format (v1–v4) with:

- * "VIBE" magic bytes for format identification
- * Gzip-compressed token sequences (bincode serialization)
- * Base64-encoded Roaring Bitmaps
- * JSON metadata (file paths, line offsets, lexicon)

Persist + reload: 4 ms (10 files), 150 μ s (100 files).

5.4 Hybrid Search

The HybridSearcher combines BM25 document-level retrieval with Vibe Index positional precision:

```
Algorithm HybridSearch(query):
  // Stage 1: BM25 candidate retrieval
  bm25_candidates = bm25.score_and_rank(query, top_k=3)

  // Stage 2: Vibe Index exact position validation
  results = empty set
  For each candidate in bm25_candidates:
    vibe_results = vibe_index.phrase_search(extract_phrases(query))
    vibe_results = filter_by_file(vibe_results, candidate.file_path)
    results = results  $\cup$  vibe_results

  return rank_by_confidence(results)
```

BM25 finds relevant documents; Vibe Index pinpoints exact lines within those documents. This hybrid pattern leverages BM25's semantic approximation capability and Vibe Index's positional precision.

5.5 Prompt Injector

The PromptInjector component bridges search results and LLM context construction:

- * Context window extraction: For each match, extracts $\pm N$ tokens around the match position (configurable)
- * Confidence filtering: Only includes matches above a confidence threshold (default 0.5)

- * Token budget management: Limits total injected tokens to prevent context window overflow
- * KV cache savings estimation: Calculates VRAM savings from reduced context size

```
// Example: 7B model, KV cache ~1.5 KB per token in context
full_chunk_vram = 4096 × 1.5 KB = 6.1 MB (entire chunk in KV cache)
minimal_context_vram = 300 × 1.5 KB = 0.45 MB (relevant context only)
per_query_savings = 6.1 MB - 0.45 MB = 5.65 MB VRAM saved per query

// Cumulative savings across a session with ~1000 queries:
// 5.65 MB × 1000 queries ≈ 5.5 GB total VRAM saved
```

6. Evaluation

6.1 Experimental Setup

All benchmarks were run on a single core in release build (`cargo bench --release`). Test corpus: 50K synthetic tokens simulating Rust source code patterns. File-aware benchmarks use 10, 50, 100, and 500 files with realistic source code content.

6.2 Phrase Search Performance

Operation	Latency
Index 50K tokens	1.83 ms
Index 10K tokens	425 µs
Exact phrase (1 match)	112 ns
Exact phrase (~100 matches)	198 µs
Phrase not found (early exit)	82 ns

The 112 ns single-match latency demonstrates the efficiency of the anchor-and-offset approach. When the anchor token has cardinality 1, I perform only 2 `bitmap contains()` checks (for the phrase length 3), each taking approximately 50 ns on modern hardware.

6.3 Fuzzy Search Performance

Operation	Latency
Fuzzy 1-char typo	4.64 µs
Fuzzy 2-char typo	67 µs
Fuzzy no match (early exit)	127 ns

The bigram prefiltering makes fuzzy search practical for interactive use. A 1-character typo is resolved in under 5 μ s, enabling real-time typo-tolerant code search.

6.4 File-Aware Search

Operation	Latency
Phrase search (10 files)	25 μ s
Phrase search (50 files)	127 μ s
Phrase search (100 files)	250 μ s
Phrase search (500 files)	1.15 ms
Add file (100 files)	7 ms
Persist + reload (10 files)	4 ms
Persist + reload (100 files)	150 μ s

File-aware search adds minimal overhead compared to raw token search, as file metadata is stored alongside position data and accessed via binary search ($O(\log n)$).

6.5 Comparison with Alternative Approaches

Approach	Precision	Latency	Memory (50K tokens)
Vibe Index	Exact token position	70 ns – 120 μ s	~0.5 MB
BM25	Document-level match	50–500 μ s	~2 MB
FAISS (embeddings)	Semantic similarity ~0.85	5–20 ms	~20 MB
Tantivy	Document-level match	50–200 μ s	~3 MB

Vibe Index is 100–1000 $\times$ faster than embedding search and provides exact positions (not document-level matches). The memory footprint is approximately 40 $\times$ smaller than FAISS.

6.6 RAG Context Optimization

Using Vibe Index for context injection in RAG pipelines:

Metric	Traditional RAG	Vibe Index RAG	Improvement
Tokens injected per query	~4,096	~300	93% reduction
KV cache VRAM (7B model)	~6.1 MB	~0.45 MB	93% reduction (5.65 MB saved per query)
Estimated inference overhead	~30+ seconds	~2 seconds	93% reduction

6.7 Integration Tests on Real Codebase

Integration tests index the Vibe Index project's own `src/` directory (2,792 lines of Rust across 10 files) and verify:

- * Phrase search on actual code patterns (`fn authenticate`, `let mut`)
- * Fuzzy search on typos in real identifiers
- * Confidence ranking correctness
- * File-level result grouping

7. LLM Integration

7.1 llama.cpp Integration

The optional `llama-cpp` module (enabled via `--features llama-cpp`) provides a complete pipeline: index → search → build prompt → llama.cpp completion via HTTP. Pre-built templates support refactoring, bug-finding, and documentation generation tasks.

7.2 vLLM Integration

The optional `vllm` module (enabled via `--features vllm`) provides production-ready integration with vLLM's OpenAI-compatible API:

- * Context budget management: Limits injected tokens to prevent context overflow
- * Output validation: Post-injection checks (token boundaries, balanced braces)
- * Sanity checks: Syntax validation of generated code
- * Confidence feedback loop: Uses output quality to adjust search parameters

7.3 MCP Server

The Python MCP server (`mcp_server.py`) exposes Vibe Index as a set of tools via the Model Context Protocol, enabling integration with LLM applications including LM Studio, Ollama, Claude Desktop, and

OpenCode. Available tools: `index_text`, `index_file`, `phrase_search`, `fuzzy_search`, `search`, `get_file_content`, `get_stats`, `clear_index`.

8. Limitations and Future Work

8.1 Current Limitations

- * No semantic search: Vibe Index performs exact matching only — "login" does not match "authenticate". Semantic matching requires embedding-based retrieval as a complementary first stage
- * BM25 IDF computed on-the-fly: Negligible for small document sets (<10K documents), but measurable at larger scale. Precomputing IDF vectors would reduce per-query latency
- * Hot layer size is immutable: ``max_hot_tokens`` is fixed at ``HotColdIndex`` creation time. Dynamic resizing would improve memory utilization
- * No SIMD optimization: I tested AVX2/AVX-512 on Roaring bitmap iteration and found 64–115% slowdown. Run-compression patterns in position bitmaps do not benefit from fixed-width SIMD operations

8.2 Future Work

- * Semantic prefiltering: Embedding-based document selection → Vibe Index exact positioning, creating a unified semantic+positional retriever
- * Distributed indexing: Sharding across multiple nodes for billion-token corpora
- * Streaming index: Incremental updates without full re-indexing for live codebases
- * Multi-language query parsing: Extended `query_parser` for Python, JavaScript, Go, and other languages
- * GPU acceleration: Exploring GPU-accelerated bitmap operations for ultra-large corpora

9. Conclusion

Vibe Index demonstrates that Roaring Bitmaps, when combined with anchor-and-offset phrase matching and bigram prefiltering for fuzzy search, provide a powerful foundation for sub-microsecond exact phrase matching. The system achieves 112 ns latency for single-match queries and 1.83 ms for indexing 50K tokens, while using only 0.5 MB of memory.

For LLM context retrieval, Vibe Index addresses a critical gap in the RAG pipeline: not just finding relevant documents, but finding the exact positions within those documents where relevant content resides. This enables 93% token reduction and estimated 5.65 MB KV cache VRAM savings per query (~5.5 GB cumulative across 1000 queries) for 7B parameter models.

Vibe Index is not a replacement for semantic search — it is a complement. The recommended pattern is: embeddings or BM25 find candidate chunks → Vibe Index pinpoints exact positions within them →

only the relevant context window is injected into the LLM prompt.

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This research was developed as an independent project. The source code is available at <https://github.com/mladenpop-oss/vibe-index> under the MIT license.